

Is this a successful proof of Fermat's Last Theorem?

by John B. van der Linden

[0] $a^n + b^n = c^n$ where $a < b < c$ and a, b, c , and n are positive integers with $n \geq 2$

since $c > a$ and $c > b$, c can be written as $x+a$ or $y+b$ where x and y are positive integers

[1] thus $a^n + b^n = (x+a)^n = (y+b)^n$

now expand $(x+a)^n$ and $(y+b)^n$:

$(x+a)^n = x^n + a^n + \text{remaining terms (call their sum "f")}$

$(y+b)^n = y^n + b^n + \text{remaining terms (call their sum "g")}$

thus $a^n + b^n = x^n + a^n + f = y^n + b^n + g$

subtract from both sides

$b^n = x^n + f$

$a^n = y^n + g$

f and g are both positive integers, thus $b^n > x^n$ and $a^n > y^n$

take n th root of both sides

$b > x$ and $a > y$

write b as $x+s$ and a as $y+r$ where s and r are positive integers

now back to [1]

$a^n + b^n = (x+a)^n = (y+b)^n$

$a^n + b^n = (x+(y+r))^n = (y+(x+s))^n$

$a^n + b^n = (x+y+r)^n = (x+y+s)^n$

thus $r = s$; rename this integer Z .

[2] then $a^n + b^n = (x+y+Z)^n$

by definition, $a \neq b$ and since $c = x+a = y+b$, $x \neq y$

[3] $y+Z = a$ and $y < a$, thus **a must be at least 2** (if $y = Z = 1$)

$b > a$ so **b must be at least 3**

[4] thus $a + b \geq 5$

now back to [2]:

$a^n + b^n = (x+y+Z)^n$

recall that $b = x+Z$ and $a = y+Z$. thus $x = b-Z$ and $y = a-Z$. Substitute these values:

$a^n + b^n = ((b-Z)+(a-Z)+Z)^n$

[5] $a^n + b^n = (a + b - 2Z + Z)^n = (a + b - Z)^n$

Suppose we imagine that $(a, b, Z) = (2, 3, 1)$ works for some unknown integer value of n .

Substituting these values of a, b , and Z into [5] gives

[6] $2^n + 3^n = (2 + 3 - 1)^n = 4^n$

Divide both sides by 4^n

$$\left(\frac{1}{2}\right)^n + \left(\frac{3}{4}\right)^n = 1$$

Both terms on the left are less than 1 and get smaller as n increases, thus n must be small.

Substituting $n = 1$ gives $\left(\frac{1}{2}\right) + \left(\frac{3}{4}\right) = 1.25$ which is greater than 1

Substituting $n = 2$ gives $\left(\frac{1}{4}\right) + \left(\frac{9}{16}\right) = 0.8125$ which is less than 1

All larger values of n will give a value for $\left[\frac{1}{4}\right]$ that is less than 0.8125, since both exponent terms are decreasing

Thus n must be between 1 and 2 if $a = 2$, $b = 3$, and $Z = 1$. But this violates our initial condition that n is a positive integer

[7] Thus $a = 2$, $b = 3$, and $Z = 1$ cannot be correct

Is Z wrong?

Recall that $b = x+Z$ and $a = y+Z$, where $c = x+a = y+b$. We know $1 \leq x$ and $1 \leq y$, so $1 \leq b - Z$ and $1 \leq a - Z$. Thus $Z \leq a - 1$ and $Z \leq b - 1$. If our current values $a = 2$ and $b = 3$ have a solution then

$$Z \leq a - 1$$

$$Z \leq 2 - 1$$

$$Z \leq 1$$

So there are no other possible values of Z for $a = 2$ and $b = 3$, and the only one available, $Z = 1$, leads to a value of n that violates initial conditions. Thus the issue lies with $a = 2$ and $b = 3$.

The next lowest possible value for (a,b) is $(3,4)$. We know this works, since $3^2 + 4^2 = 5^2$.

[8] Therefore, **$a \geq 3$ and $b \geq 4$, and $a + b \geq 7$.**

What about Z ? For the Pythagorean solution $3^2 + 4^2 = 5^2$, $Z = (3+4) - 5 = 2$. Can Z ever be smaller than this? That is, can Z ever equal 1?

We know $a + b \geq 7$. Subtract 1 from both sides

$$a + b - 1 \geq 7 - 1$$

Raise to the power of n

$$(a + b - 1)^n \geq 6^n$$

[9] Imagine that $Z = 1$. Then the left hand side can be written as $(a + b - Z)^n$.

$$(a + b - Z)^n \geq 6^n$$

Substitute from [5]:

$$a^n + b^n \geq 6^n$$

We know $b > a$ and $b^n > a^n$, so we can write

$$2 \cdot b^n \geq a^n + b^n \geq 6^n, \text{ or simply}$$

$$2 \cdot b^n \geq 6^n$$

Take n th root of both sides

$$2^{(1/n)*b} \geq 6$$

$$2^{(1/n)} \geq 6 / b$$

The largest possible value of $6 / b$ is $6 / 4 = 1.5$. So

$$2^{(1/n)} \geq 6 / 4 \geq 6 / b$$

$$2^{(1/n)} \geq 1.5$$

But for $n \geq 2$, the maximum value of $2^{(1/n)}$ is $1.4142... \geq 1.5$

Thus the initial assumption of $Z = 1$ from [9] cannot be correct.

[10] So **Z must be at least 2**

Now return to [5].

$$a^n + b^n \leq (a + b - Z)^n$$

Z must be at least 2.

$$a^n + b^n \leq (a + b - 2)^n$$

We know the minimum value of the left side is $3^2 + 4^2 = 5^2 = 25$.

$$[11] \text{ Thus } 25 \leq a^n + b^n \leq (a + b - 2)^n$$

Substitute $k = a+b$

$$25 \leq (k-b)^n + (k-a)^n \leq (k-2)^n$$

Since $b > a$, we know $(k-b)^n < (k-a)^n$. Thus we can write

$$25 \leq 2^*(k-b)^n \leq (k-b)^n + (k-a)^n \leq (k-2)^n$$

Or simply

$$[12] \text{ } 25 \leq 2^*(k-b)^n \leq (k-2)^n$$

Write the factor 2 (from the middle part) as $(2^{(1/n)})^n$

$$25 \leq (2^{(1/n)})^n * (k-b)^n \leq (k-2)^n$$

Take the nth root of both sides

$$25^{(1/n)} \leq 2^{(1-n)} * (k-b) \leq k - 2$$

Divide by $(k-b)$

$$(25^{(1/n)}) / (k - b) \leq 2^{(1-n)} \leq (k - 2) / (k - b)$$

Back-substitute $a + b = k$

$$(25^{(1/n)}) / a \leq 2^{(1-n)} \leq (a + b - 2) / a$$

$$[13] \text{ } (25^{(1/n)}) / a \leq 2^{(1-n)} \leq 1 + (b/a) - (2/a)$$

The largest possible value of $(25^{(1/n)}) / a$ is at $a = 3$. So

$$(25^{(1/n)}) / a \leq (25^{(1/n)}) / 3 \leq 2^{(1-n)} \leq 1 + (b/a) - (2/a)$$

Ignore the rightmost part.

$$[14] \text{ } (25^{(1/n)}) / a \leq (25^{(1/n)}) / 3 \leq 2^{(1-n)}$$

Raise everything to the power of n.

$$[15] \text{ } 25 / (a^n) \leq 25 / (3^n) \leq 2$$

Ignore the leftmost part.

$$\lceil 16 \rceil \quad 25 / (3^n) \leq 2$$

$$3^n / 25 \leq 1/2$$

$$3^n \leq 25/2$$

$$n \leq \log_{\text{base } 3} \text{ of } 12.5$$

$$n \leq 2.2990\dots$$

$\lceil 17 \rceil$ The only possible integer value of n is 2.

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Note: I was stuck at $\lceil 6 \rceil$, and decided to ask Google's AI Overview tool, "do the graphs of $f(n) = (2^n + 3^n)$ and $g(n) = 4^n$ intersect anywhere?" It stated yes, at only one point ($n = 1.507\dots$) (I failed to specify that n needed to be an integer). It then provided the technique of dividing both sides by 4^n , noting the fact that both left-hand terms decreased, and evaluating the equation for $n = 1$ and $n = 2$.

The rest of the proof is my own original work.